

Design vs Implementation Report

Shop Website System

1. Introduction

This report evaluates the relationship between the original design provided in “**Shop Website.pdf**” and the current implemented system. The purpose is to demonstrate how the initial design intent has been preserved, what enhancements have been introduced, and where intentional deviations were made during development.

The original design included seven core screens: authentication, homepage, product view, cart/checkout, and user profile. The current system is implemented as a modern web application using a React-based frontend and a backend API with database integration.

2. Design Source and Implementation Overview

- **Design Source:** *Shop Website.pdf* (wireframes covering authentication, navigation, product display, cart, and profile)
- **Implementation Stack:**
 - Frontend: React + Vite (Single Page Application)
 - Backend: Hono API
 - Database: PostgreSQL with Prisma ORM
 - Routing: Defined in `src/App.tsx`

3. Alignment with Original Design

The current system maintains strong alignment with the original design in terms of structure, user flow, and branding.

3.1 Branding and Visual Identity

The original hero message “**Beauty Inspired by Real Life**” has been preserved on the homepage. While supporting text has been slightly refined, the overall brand positioning remains consistent with the initial concept.

The site title “**SHOP**” is still used in the navigation bar, maintaining continuity with the original design identity.

3.2 Navigation and Information Architecture

The global navigation structure from the design—**Home, Products, About, Contact, and Sales**—is fully implemented. The application introduces an additional **Admin** section, which is conditionally displayed for administrative users.

This ensures that the original information architecture is retained while supporting extended functionality.

3.3 Authentication System

The design included **Sign Up** and **Login** functionality using basic credentials. The implementation expands on this by requiring:

- Name
- Email
- Password
- Phone number

This adjustment improves usability and aligns better with real-world e-commerce requirements.

3.4 User Profile

The profile section in the design includes personal details such as name, email, and phone number. This is fully implemented in the /profile route, using a reusable layout structure for consistency across user pages.

3.5 Product and Shopping Flow

The product page design—featuring image, price (in Namibian Dollars), rating, discount messaging, and a **BUY NOW** action—is accurately reflected in the implementation.

The shopping flow remains unchanged:

- Browse products
- View product details
- Add to cart
- Proceed to checkout

Cart functionality includes item quantities, totals, and a checkout process consistent with the original design.

3.6 Footer and Contact Section

The footer retains the structure from the design, including:

- “Keep in Touch” section
- Customer service information

- Contact details

A dedicated **Contact page** is also implemented, maintaining the communication intent of the original design.

4. Enhancements Beyond the Original Design

Several features were added during development to transform the static design into a functional, production-ready system.

4.1 Backend and Database Integration

The system includes a fully functional backend API and database:

- REST API built with Hono
- PostgreSQL database
- Prisma ORM for data management

This enables real-time data handling for users, products, and orders.

4.2 Administrative Interface

An **Admin panel** (/admin/*) has been introduced to manage:

- Products
- Orders
- Users
- Basic analytics

This functionality was not present in the original design but is essential for system operation.

4.3 Authentication and Security

Security features have been implemented, including:

- JWT-based authentication
- Session validation
- Automatic logout on inactivity

These ensure data integrity and user security.

4.4 WhatsApp Integration

Instead of traditional email-based communication, the system uses **WhatsApp** for:

- Order confirmations

- Customer inquiries

This aligns with practical business requirements and improves accessibility.

4.5 Deployment Readiness

The system is designed for deployment using platforms such as:

- Vercel
- Render
- Cloudflare Pages

This ensures scalability and real-world usability.

5. Intentional Differences from the Original Design

Some deviations were necessary due to scope, usability improvements, or implementation constraints.

5.1 Sidebar Structure

The original design included items such as:

- More
- Orders
- History
- Messages

The implementation replaces these with:

- Profile
- Home
- Products
- About
- Contact
- Sales

Order-related functionality is integrated within the profile and checkout process rather than separate pages.

5.2 Contact Method

The design suggested a form-based submission system. This has been replaced with a **WhatsApp-based communication flow**, maintaining the same objective but using a more practical channel.

5.3 Password Recovery

The **“Forgot Password”** feature shown in the design has not been implemented due to scope limitations. This would require additional infrastructure such as email services.

5.4 Content and Data

Placeholder text (e.g., Lorem Ipsum) used in the design has been replaced with dynamic product data stored in the database. Sample user identities from the design are not retained.

6. Conclusion

The implemented system successfully preserves the **core design intent** of the original mockups. The overall user journey—browsing products, managing a cart, checking out, and maintaining a user profile—remains unchanged.

While the system is not a pixel-perfect replica of the design, all deviations are intentional and justified. Most differences arise from:

- Enhancing functionality
- Meeting real-world business requirements
- Implementing technical infrastructure

Rather than replacing the original design, the current system **extends and operationalizes it**, transforming static wireframes into a fully functional application.

7. Repository References

- Routing Configuration: `src/App.tsx`
- Architecture Documentation:
 - ARCHITECTURE STRUCTURE
- Technology Stack: TECH STACK DOCUMENT